

Why S-X2 rejects too often under a true Rasch model

Null diagnostic of `mirt::itemfit(fit_stats = "S_X2")`

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Purpose

In an earlier comparison of S-X2 with `infit` (July 2026, since replaced by `sx2_comparison_v2.qmd`), uncorrected S-X2 from `mirt::itemfit()` rejected 6.9 to 8.8 percent of items under a true Rasch model, against a nominal 5, and BH over 20 items flagged at least one item in 10.5 to 19 percent of datasets, rising with n . This note finds the cause. The design is in `sx2-null-diagnostic-design.md` and the simulation in `sx2_null_diagnostic.R`.

`mirt` computes the degrees of freedom for each item as the number of score groups after collapsing, minus the item's own parameters, minus every estimated latent parameter. For `itemtype = "Rasch"` the latent variance is estimated, so each item loses one df that Orlando and Thissen's definition does not remove. The deduction is deliberate. `mirt`'s NEWS lists it under version 1.20.1 as a df adjustment "for models where the latent trait hyper-parameters have been estimated". If S-X2 follows a chi-square on $df + 1$ and is referred to df , the rejection rate at .05 is 6.9 to 7.5 percent at the df values seen here.

Design

Null only, with fresh data for every replication. Three item sets: 20 dichotomous items evenly spaced on $[-2, 2]$, the 20 dichotomous locations of the original study, and 9 items with four categories on $[-1.5, 1.5]$. Persons from $N(0, 1.5^2)$, $n = 150, 500$ and 2000 , 2000 replications per cell. The Monte Carlo SE is about 0.5 points for a 5 percent rate, per item and family-wise.

Variant	Fit	Reference
V0	Rasch, variance estimated	<code>mirt</code> default
V1	same as V0	p recomputed on $df + 1$
V2	same as V0, <code>mincell = 5</code>	<code>mirt</code> default
V2c	same as V2	p recomputed on $df + 1$
V3	Rasch, variance fixed at the true 2.25	<code>mirt</code> default, which then has no latent deduction

V1 tests the df. V3 removes the deduction and the estimation error in the variance together, so V1 against V3 separates the two. V2 tests stricter collapsing of sparse cells.

```
ok <- !vapply(sim$res, is.null, logical(1))
long <- do.call(rbind, lapply(which(ok), function(i) {
  g <- sim$grid[i, ]; m <- sim$res[[i]]
  data.frame(set = g$set, n = g$n, rep = g$rep, item = seq_len(nrow(m)), m)
})) |>
mutate(p_v1 = pchisq(S_v0, df_v0 + 1, lower.tail = FALSE),
       p_v2c = pchisq(S_v2, df_v2 + 1, lower.tail = FALSE))

pv <- long |>
select(set, n, rep, item, V0 = p_v0, V1 = p_v1, V2 = p_v2, V2c = p_v2c,
       V3 = p_v3) |>
pivot_longer(V0:V3, names_to = "variant", values_to = "p")

wide <- function(d, value) d |>
  pivot_wider(names_from = n, values_from = all_of(value), names_prefix = "n = ")
```

No replication was discarded. elapsed is 7.3 minutes on 10 cores, mirt 1.46.1.

Results

The statistic is chi-square on df + 1

If the reference is right, the mean of S-X2 equals its df and the variance is twice the df.

```
long |>
group_by(set, n) |>
summarise(`mean S - df, V0` = mean(S_v0 - df_v0, na.rm = TRUE),
          `mean S - df, V3` = mean(S_v3 - df_v3, na.rm = TRUE),
          `var S / 2(df + 1), V0` = var(S_v0 - df_v0, na.rm = TRUE) /
            (2 * mean(df_v0 + 1, na.rm = TRUE)),
          `mean df, V0` = mean(df_v0, na.rm = TRUE), .groups = "drop") |>
kable(digits = 2)
```

set	n	mean S - df, V0	mean S - df, V3	var S / 2(df + 1), V0	mean df, V0
dich_even	150	1.07	0.07	0.99	9.85
dich_even	500	0.98	-0.02	1.03	13.76
dich_even	2000	0.96	-0.04	1.03	15.87
dich_orig	150	1.02	0.02	0.99	9.35
dich_orig	500	1.03	0.04	1.04	13.25
dich_orig	2000	0.99	-0.01	1.03	15.67
poly	150	1.66	0.64	0.90	20.82
poly	500	1.08	0.09	1.01	36.19
poly	2000	0.86	-0.11	1.02	45.65

Under mirt's df, S-X2 exceeds its df by 0.96 to 1.07 on average for dichotomous items, and its variance matches $2(df + 1)$ to within 5 percent from $n = 500$. With the variance fixed, which takes away the deduction, the mean excess is within 0.07 of zero. The polytomous set shows the same pattern from $n = 500$. At $n = 150$ its excess is 1.66 under V0 and 0.64 under V3, so something else adds to the statistic there.

Per-item rejection

```
pv |>
  group_by(set, variant, n) |>
  summarise(rate = 100 * mean(p < .05, na.rm = TRUE), .groups = "drop") |>
  wide("rate") |>
  kable(digits = 2)
```

set	variant	n = 150	n = 500	n = 2000
dich_even	V0	7.53	7.07	7.11
dich_even	V1	4.99	5.09	5.23
dich_even	V2	10.46	7.35	6.86
dich_even	V2c	4.54	4.69	4.96
dich_even	V3	4.99	5.12	5.20
dich_orig	V0	7.19	7.35	7.16
dich_orig	V1	4.76	5.26	5.17
dich_orig	V2	10.47	7.59	7.07
dich_orig	V2c	4.41	4.84	4.89
dich_orig	V3	4.76	5.29	5.17
poly	V0	6.79	6.18	5.95
poly	V1	4.99	5.03	4.83
poly	V2	14.79	8.47	5.99
poly	V2c	7.58	6.16	4.79
poly	V3	4.81	5.06	4.84

Adding the missing df brings every cell to nominal: V1 rejects 4.8 to 5.3 percent of items against 6.0 to 7.5 under V0. V1 and V3 agree to within 0.2 points in every cell, so estimating the latent variance costs nothing. The df is the whole of the per-item excess.

Stricter collapsing on its own (V2) is worse than the default, because `mincell = 5` merges so many score groups at $n = 150$ that the df falls to about 2, and a one-df shortfall weighs more at small df. With the df corrected (V2c) it is close to nominal for dichotomous items, but at $n = 150$ it leaves too few score groups to say much. For dichotomous items 239 to 278 of the 40,000 item-replications at $n = 150$ had no df left.

The far tail and the family-wise rate

BH over 20 items rejects on p-values near $.05/20$, so the family-wise rate depends on the far tail, not on the 5 percent point.

```
pv |>
  group_by(set, variant, n) |>
  summarise(`per 1000` = 1000 * mean(p < .005, na.rm = TRUE), .groups = "drop") |>
  wide("per 1000") |>
  kable(digits = 1, caption = "Items with p < .005, per 1000 (nominal 5)")
```

set	variant	n = 150	n = 500	n = 2000
dich_even	V0	10.3	10.9	9.8

set	variant	n = 150	n = 500	n = 2000
dich_even	V1	6.3	7.7	6.8
dich_even	V2	11.3	7.6	8.1
dich_even	V2c	3.6	4.5	5.3
dich_even	V3	6.5	7.7	6.8
dich_orig	V0	10.8	10.3	9.3
dich_orig	V1	6.8	6.9	6.4
dich_orig	V2	11.9	8.6	8.4
dich_orig	V2c	3.4	5.1	5.2
dich_orig	V3	6.7	6.8	6.5
poly	V0	5.9	8.4	7.5
poly	V1	4.2	6.8	5.8
poly	V2	17.9	9.1	7.8
poly	V2c	6.9	6.0	5.9
poly	V3	4.0	6.8	5.6

```
pv |>
  group_by(set, variant, n, rep) |>
  summarise(bh = any(p.adjust(p, "BH") < .05, na.rm = TRUE), .groups = "drop") |>
  group_by(set, variant, n) |>
  summarise(fw = 100 * mean(bh), .groups = "drop") |>
  wide("fw") |>
  kable(digits = 2, caption = "Datasets with at least one BH-flagged item (%)")
```

set	variant	n = 150	n = 500	n = 2000
dich_even	V0	11.90	12.50	10.75
dich_even	V1	7.10	8.35	6.95
dich_even	V2	11.45	8.40	9.15
dich_even	V2c	3.55	5.15	5.85
dich_even	V3	7.30	8.55	6.80
dich_orig	V0	11.40	11.45	10.70
dich_orig	V1	7.45	8.55	6.75
dich_orig	V2	11.55	9.50	9.40
dich_orig	V2c	3.90	5.00	5.65
dich_orig	V3	7.20	8.70	6.95
poly	V0	6.00	7.95	7.00
poly	V1	4.30	6.45	5.60
poly	V2	17.00	9.40	7.85
poly	V2c	7.10	6.00	5.55
poly	V3	3.95	6.40	5.70

Correcting the df leaves part of the problem. Under V1, items reach $p < .005$ at 6.4 to 7.7 per 1000 for dichotomous items against a nominal 5, and BH flags at least one item in 6.8 to 8.6 percent of datasets. That is down from 10.7 to 12.5 under V0, but still above 5. The polytomous set is close to nominal under V1, at 4.3 to 6.5 percent. Under V0, V1 and V3, Bonferroni gives the same rates to within 0.3 points, as expected under a complete null. V2c removes the tail excess for dichotomous items (3.4 to 5.3 per 1000 and 3.6 to 5.9 percent family-wise), which points to sparse score groups as the source of what remains.

No rise with sample size

Under V0 the family-wise rate is 10.7 to 12.5 percent at every n for both dichotomous sets, including the original locations. The rise to 19 percent at $n = 2000$ in the original study does not reproduce with fresh data. There, every replication was drawn from one master dataset of 10,000 respondents. Each such sample is, taken on its own, a random sample from the model, so the design does not bias the rates. It makes the replications dependent instead. At $n = 2000$ the samples share about a fifth of their respondents, so any chance departure from the model in the master dataset appears in every replication, and the rate in a cell varies between master datasets by more than its Monte Carlo SE allows. The rise is most likely such a draw, and the same holds for the other methods in that study and for its per-item spread of 4.4 to 11.7 percent. Here, under V1 at $n = 2000$, per-item rates run from 4.0 to 6.0 percent, which is about what 2000 independent replications give by chance.

Interpretation

1. The per-item excess is mirt's df. Removing the latent variance from every item's df makes S-X2 reject 6 to 7.5 percent of items at a nominal 5 under a true Rasch model. Adding the df back, or fixing the variance, gives 4.8 to 5.3. Whether the adjustment is right for other models is not tested here, but for the Rasch and partial credit models it makes the test liberal.

2. A smaller tail excess remains, from sparse cells. With the df corrected, 20 dichotomous items still give 6.8 to 8.6 percent family-wise under BH. Stricter collapsing removes it, at the cost of most of the score groups at small n . Nine polytomous items are close to nominal.

3. The rise with n in the original study is most likely noise from its design. Resampling one master dataset makes its replications dependent, so differences between sample sizes there are less certain than they look. With fresh data the rate does not rise with n .

For the revised comparison. S-X2 should enter with the corrected df. Since the corrected test is still about 2 to 3.5 points over 5 percent family-wise for 20 dichotomous items, it should also be reported against its own null in the same run, or given a simulation-based cutoff, so that it is not compared with family-wise controlled methods at a higher error rate. Fresh data per replication is needed for every method.

Whether to raise the df adjustment with the mirt maintainer is a separate decision.

Reference

Orlando, M., & Thissen, D. (2000). Likelihood-based item-fit indices for dichotomous item response theory models. *Applied Psychological Measurement*, 24(1), 50-64.